Definition of γ_j

Given a time series of real numbers $x(n)$, form d-tuples:

$$v(i) = (x(i), x(i-1), ..., x(i-d+1)) = (v_1(i), v_2(i), ..., v_d(i)).$$

Define an indicator function $I_{ij}(\epsilon)$: 1 if each component of vector $v(i)$ is within ϵ of the corresponding component of vector $v(j)$, 0 otherwise.

$$I_{ij}(\epsilon) = \prod_{k=1}^{d} \Theta(\epsilon - |v_k(i) - v_k(j)|)$$

Define $C_d(\epsilon) = \frac{1}{N} \sum_{pairs} I_{ij}(\epsilon)$, where N is the number of pairs of vectors $(v(i), v(j))$.

$C_d(\epsilon)$ measures the probability that two d-vectors are within ϵ of each other in all their Cartesian coordinates.

Define

$$\gamma_j = 1 - \frac{C_j^2}{C_{j-1} C_{j+1}}.$$

γ_j measures the degree of additional dependence of $x(i)$ on $x(i-j)$, aside from the trivial one induced by the explicit dependencies on $x(k)$ for $i - j < k < i$.